

Teacher Rubric Reference

Raw synthetic teacher material - separate from student submissions

Q1 - 5 points

- 1: chooses $u=x^2$
- 2: carries the $1/2$ factor
- 2: transforms bounds and evaluates $(e-1)/2$

Q2 - 8 points

- 2: resolves gravity and normal force
- 2: friction opposes motion
- 2: a is approximately 3.20 m/s^2
- 2: v is approximately 4.38 m/s with units

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Q3 - 7 points

- 2: proves $\ker(A) \subset \ker(A^T A)$
- 3: uses $x^T A^T A x = \|Ax\|^2$ for the reverse inclusion
- 2: concludes equal rank by rank-nullity

Q4 - 10 points

- 2: returns `[]` for empty input
- 3: subtracts `max(xs)`
- 2: normalizes correctly
- 3: passes large-value tests